A workout plan for shift workers: training around rotating hours

A rota doesn't care which day it is. A plan built on Monday, Wednesday and Friday breaks the first week the hours move. Here's a plan counted per rota, not per calendar: two or three full-body sessions anchored to your shifts, a schedule for days, nights and twelve-hour patterns, and a floor for the long runs.

> **Short answer:** Train two or three times per rota week, full body, and anchor each session to a shift instead of a clock: after you wake and before you leave for a day or night shift, or on your days off after a run of twelve-hour shifts. Plan the sessions the day the rota comes out. On a long run, one hard session at your usual weights is the floor. Which hour you train makes little difference to strength or muscle in the research, so train in the one you can keep.

By Diogo · Updated October 31, 2026 · 9 min read

Almost every workout plan is written on a calendar: three days a week, Monday, Wednesday, Friday, same hour each time. Shift work breaks that in the first week. The days move, the hours move, and a run of nights or twelve-hour shifts can swallow a whole week's sessions. Then the plan says you failed, when it's the plan that didn't fit.

This plan is counted per rota, not per calendar week. It's two or three full-body sessions anchored to your shifts, a schedule for the common patterns, and a floor for the stretches when the rota wins. It's about where the sessions go. Questions about sleep and health on shift work are for a doctor; this page is training only. For the research on morning versus evening lifting, the best time to work out is the page; this one assumes you've read that your best time is the one you can keep.

The idea: anchor sessions to the shift, not the clock

Three findings make a shift plan possible.

- **The hour matters less than you'd think.** A 2019 meta-analysis of 11 studies found strength gains were similar whether people trained in the morning or the evening, and so were gains in muscle size.[1] One honest caveat: the trials in a 2023 review of training time all ran on day schedules, with morning sessions at about 8 a.m. and later sessions at about 5 p.m. on average, all between 7:30 a.m. and 7 p.m.[2] None tested lifting at 3 a.m. between night shifts, so the night-shift parts of this page are practical scheduling, not research.
- **Which days don't matter much either.** A meta-analysis of 25 studies found that when the total amount of training was matched, how many days it was spread over made no meaningful difference to muscle growth, and concluded people can choose how often to train a muscle "based on personal preference".[3] Two sessions on consecutive days off, then nothing for four, is a legitimate week.

- **Two sessions a week is the dose that counts.** The American College of Sports Medicine's 2026 position stand makes lifting with high effort at least twice a week, with all major muscle groups trained, its primary recommendation, and says even "minimal doses" of lifting bring substantial gains.[4] Two or more days a week is also the World Health Organization's guideline for muscle-strengthening activity.[5]

So the rule is: the session is attached to an event that happens on every shift pattern, like "after I wake up, before I leave", or to a day off, never to "6 p.m. on Wednesday". Then the plan moves with the rota instead of fighting it.

The plan: two full-body sessions, alternated

Two sessions, A and B, alternated: whichever you did last, do the other next. Each has five slots in the same order (a squat, a hinge, a push, a pull and a core move), so every session trains the whole body and no session depends on the one before it. The gym version comes first; the home version follows, for the days the gym's hours don't match yours.

Slot	Day A	Day B
Squat	Leg press or goblet squat. Home: goblet squat with one dumbbell, or a slow bodyweight squat	Hack squat or split squat. Home: reverse lunge
Hinge	Romanian deadlift. Home: the same with dumbbells, or a single-leg hip bridge	Hip thrust. Home: glute bridge with a dumbbell on your hips
Push	Dumbbell bench press or a chest-press machine. Home: push-up	Overhead dumbbell press. Home: the same, or a pike push-up
Pull	Lat pulldown. Home: one-arm dumbbell row	Seated cable row. Home: band row with the band in a door
Core	Plank	Dead bug (swaps on the core workout)

Sets and reps: three sets of eight to twelve reps, two sets in your first two weeks. With light dumbbells or bodyweight, work in twelve to twenty. Every set stops with two or three reps still in you. (Here's how to judge reps left.) Rest about ninety seconds, or pair the push with the pull and the hinge with the core move to save time: a 2021 review of time-efficient training found that supersets, drop sets and rest-pause training roughly halve a session's length while keeping the volume, and advises keeping warm-ups specific to the exercise rather than long and general.[6] (How to pair exercises.) Pair the machine or dumbbell versions, not a heavy barbell lift; the review advises against pairing those.[6] A session runs thirty-five to forty-five minutes; the paired version, about thirty.

Progression: whenever you reach the top of the rep range with reps to spare, add the smallest step next session: one notch of weight, or one rep. The ACSM's 2009 position stand on progression puts the step at 2 to 10 percent more weight once you can do one or two reps beyond your target.[7] (Progressive overload, explained.) Add a third session in weeks when the rota has room; two is the plan, three is a good week.

The schedule, by shift pattern

The day the rota comes out, put the sessions in before anything else fills the gaps, the way a student blocks training around a new term's classes. Here's where they go for the common patterns.

Pattern	Where the sessions go
Day shifts, eight hours	Before work or straight after, the same anchor every working day you train. Two sessions on work days, a third on a day off if you want it.
Evening shifts	After you wake, before you leave. The gym is quiet then. Two or three a week.
Night shifts, a block of them	After you wake and before you leave for the shift, on the first and third nights of the block. Not straight after a night shift: that's when the day's sleep goes. If the block is long, the floor below applies.
Twelve-hour shifts (three or four on, then off)	Train on the days off, not the work days: two sessions, a day apart if you can, back to back if you can't. The work days are for walking and sleeping.
Rotating days to nights	Anchor to "after I wake" on every shift and it doesn't matter which it is. On the switchover day, skip or walk; take the session the next day.
Split shifts	The gap in the middle is the slot, with the paired version so it fits in thirty minutes.

Two things make this hold. First, count sessions per rota week, whatever the rota week is: if your pattern repeats every four days, "two a week" means two per cycle; on a longer cycle, keep two per seven days where you can. Second, if the gym's opening hours don't match yours, use the home column; the ACSM stand lists home-based lifting among the kinds that improve strength and found muscle growth didn't depend on whether the loads were light or very heavy.[4] The dumbbell plan is a fuller home program, and a 24-hour gym changes the whole question.

Nurses on three or four twelve-hour shifts a week

A nursing rota of three or four twelve-hour shifts is the twelve-hour row above: lift on the days off, not the shift days. The shifts already add up to thousands of steps, and steps aren't lifting. In a pedometer study of intensive care and anaesthesia nurses in Poland, the median was about 7,400 steps over a twelve-hour shift, and about five kilometres on day shifts.[8] In 20 nurses on a perioperative intensive care unit in Slovenia, a twelve-hour weekend day shift averaged 9,003 steps (6.5 km).[9] None of that is a strength session, so the plan doesn't change: Day A and Day B on two of the days off, a third session when four days off come together, and the floor below when the rota stacks a long run of shifts. The home column works on a day off as well as the gym does. (Rotating routes instead of wards? The truck driver plan is built for no gym at all.)

The floor: a long run of shifts

Some rotas hand you seven nights in a row, or a run of twelve-hour days, and the sessions aren't going to happen. The floor for that stretch is one hard session: either day, every slot, one or two sets each, at the weights you were using. Not lighter; less. A 2021 review of the least training needed to hold on to performance found strength and muscle size could be maintained for up to 32 weeks, at least in younger

people, with as little as one lifting session a week and one set per exercise, as long as the load stayed heavy for the person lifting it; its main conclusion is that intensity, meaning how heavy the weight is for you, is the key to holding on, even when how often and how much you train drop a lot.[10]

If even one session won't fit, walk. The WHO guidelines ask adults for 150 to 300 minutes of moderate aerobic activity a week and say some physical activity is better than none.[5] A walk after a shift counts toward that. When the run ends, go back to two or three sessions at the weights you left off at, not back to week one.

Five rules for a rota

- **Plan on rota day.** The sessions go into the calendar the moment the shifts do.
- **Anchor to the shift, never the clock.** "After I wake, before I leave" survives every pattern.
- **Count per rota week.** Two is the plan, three is a good week, one is the floor on a long run.
- **Write every session down.** A 2016 meta-analysis of 138 studies, about goals in general rather than lifting, found that monitoring progress helped people reach their goals, and helped more when the progress was physically recorded.[11] When the sessions are nine days apart, the logbook is the only thing that remembers the weights. (A workout log you can copy.)
- **If it hurts, stop and ask.** Sharp pain is not soreness. If you have an injury, a medical condition or a medical question, talk to a doctor before you start. I'm a fitness coach, not a doctor.

Common questions

Should I work out before or after a night shift?

As a scheduling question: before, after you've slept. After a night shift is when the day's sleep happens, and none of the trials in the 2023 review tested lifting there.[2] If before doesn't fit, the days off do. Anything about sleep itself is for a doctor.

Is it bad to train at night?

Not for strength or muscle in the studies: morning and evening lifting built about the same.[1] The trials ran 7:30 a.m. to 7 p.m., so late night is untested rather than bad.[2] The best time to work out has the full research.

What's the best workout schedule for 12-hour night shifts?

Two sessions on the days off, the work days for sleep and a walk, and the floor session when a block runs long. The twelve-hour row in the table is written for exactly that.

Can I do this with no gym?

Yes: the home column of every slot. The bodyweight workout needs nothing at all, and the hotel workout covers the weeks you're posted away.

If you'd rather not plan it alone

A plan on a web page can't see your rota. Coaching is the version where your workouts live in the Trainerize app, built around the days you actually have, and a weekly call looks at the week you just had and rebuilds the next one around what's coming: a double shift, a block of nights, a rota that changed on Thursday. Coaching for a week that doesn't hold still explains how that works, and apply tells me about your week. Otherwise: open this week's rota, put two sessions in, and do Day A.

General education only, not medical advice. If you have a medical condition, are pregnant, or take medication, check with your doctor before changing how you train or eat.

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